Complex Number 10

1. Determine the arcs specified by the following equations. Sketch each one, showing the centre and radius of the associated circle.

   a. $\arg\left(\frac{z-2}{z}\right)=\frac{\pi}{2}$

   b. $\arg\left(\frac{z-1+i}{z-1-i}\right)=\frac{\pi}{2}$

   c. $\arg\left(\frac{z-i}{z+i}\right)=\frac{\pi}{4}$

   d. $\arg\left(\frac{z+1}{z-3}\right)=\frac{\pi}{3}$

   e. $\arg\left(\frac{z-2i}{(z+2i)}\right)=\frac{\pi}{6}$

   f. $\arg\left(\frac{z}{z+4}\right)=\frac{3\pi}{4}$

2. By the putting $z=x+iy$ or otherwise, determine the locus specified by:

   a. $Im(z)=|z|$

   b. $Re\left(1-\frac{4}{z}\right)=0$

   c. $Re\left(z-\frac{1}{z}\right)=0$

3. What is the maximum value of $|z|$ for $|z-1-i|\leq 2$

4. Show that if $|\omega|=1$ and $Re\,(\omega)=-\frac{1}{2}$, then $\omega^3=1$

5. Prove that $\frac{1+\sin\theta+i\cos\theta}{1+\sin\theta-i\cos\theta}=\sin\theta+i\cos\theta$ and deduce that $\left[1+\sin\frac{\pi}{5}+i\cos\frac{\pi}{5}\right]^5+i\left[1+\sin\frac{\pi}{5}-i\cos\frac{\pi}{5}\right]^5=0$

6. Express $\left[1+i\tan\frac{4k+1}{4m}\pi\right]^m$ in the form $a+ib$, where m and k are integers.

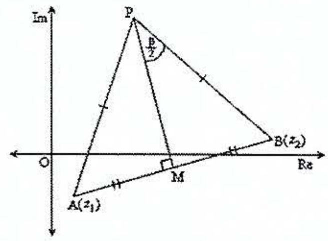
7. a. Show that $z^5+1=(z+1)(1-z+z^2-z^3+z^4)$

   b. Find in the form $\mathrm{cis}\theta$, the five complex solutions to $z^5+1=0$.

   c. If w is a complex solution of $z^5+1=0$ and $z\neq -1$, prove that $1+w^2+w^4=w+w^3$.

   d. Hence show that $\cos\frac{\pi}{5}+\cos\frac{3\pi}{5}-\frac{1}{2}=0$.

8. The diagram shows an isosceles triangle PAB. PM is the perpendicular bisector of $\angle APB$, which is β. PM bisects AB. A and B represent the complex numbers z_1 and z_2 respectively.



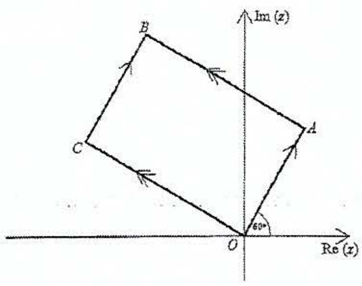
   a. Find the complex number represented by

      i. $\overrightarrow{AM}$

      ii. $\overrightarrow{MP}$

   b. Hence show that P represents the complex number $\frac{1}{2}\left(1-i\cot\frac{\beta}{2}\right)z_1+\frac{1}{2}\left(1+i\cot\frac{\beta}{2}\right)z_2$.

9. The figure below shows a parallelogram OABC in an Argand diagram. $|OA|=2$ and OA makes an angle of 60° with the positive real axis. Let z_1, z_2 and z_3 be the complex numbers represented by vertices A, B and C respectively. It is given that $z_3=(\sqrt{3}i)z_1$.



   a. Find z_1 and z_3 in the form $x+iy$.

   b. Show that $\frac{z_2}{z_1}=1+\sqrt{3}i$.

   c. Let $\omega=\cos\theta+i\sin\theta$, where $0°\leq\theta\leq 360°$. Point E is a point on the Argand diagram representing the complex number ωz_3. Find the value(s) of θ in each of the following cases:

      i. E represents the complex number z_3;

      ii. Points E, O and A lie on the same straight line.

10. If n is any integer, prove that

    a. $\left[1+\cos\frac{2\pi}{n}+i\sin\frac{2\pi}{n}\right]^n=-2^n\cos^n\frac{\pi}{n}$.

    b. $\left(1+i\sqrt{3}\right)^{2n}+\left(1-i\sqrt{3}\right)^{2n}$ has either the value 2^{2n+1} or -2^{2n}, according as to whether n is or is not a multiple of 3.

11. a. Calculate the modulus and argument of the

    i. Sum of the roots

    ii. Product of the roots of the equation $(4+3i)z^2-(3-2i)z+5+2i=0$

    b. Show that the point representing $\sin\frac{\pi}{3}+i\cos\frac{\pi}{3}$ in the Argand diagram lies on the circle of radius 1 and the centre at $(\sqrt{3},0)$.

    c. If t is real and $z=\frac{2+it}{2-it}$ show that as t varies the locus of z is a circle. Find the radius and the centre.

12. a. Draw the sketch (on an Argand diagram) of the region in which z lies if both $|\mathrm{z}-(2+\mathrm{i})|\le 4$ and $\frac{\pi}{6}\le\arg(\mathrm{z}-1-\mathrm{i})\le\frac{\pi}{2}$ are satisfied.

    b. The complex numbers $\mathrm{z}_1=-\sqrt{3}-\mathrm{i}$, $\mathrm{z}_2=\sqrt{3}-\mathrm{i}$ and $\mathrm{z}_3=2\mathrm{i}$ are represented by the points P, Q and R respectively. Show that the triangle PQR is an equilateral triangle.

13. a. By using De Moivre's Theorem or otherwise, show that $\tan 4\theta=\frac{4\tan\theta-4\tan^3\theta}{1-6\tan^2\theta+\tan^4\theta}$.

    b. Hence solve the equation $\mathrm{x}^4+4\mathrm{x}^3-6\mathrm{x}^2-4\mathrm{x}+1=0$. Give the solutions in exact form.

14. Describe the locus in the Argand diagram of the point P which represents the complex number z where $z\bar{z}-4(z+\bar{z})=9$.

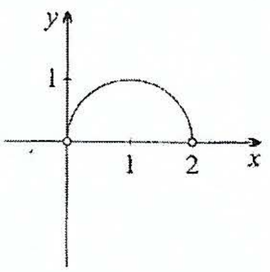
15. Let $z=x+iy$ be a complex number, where x and y are real numbers. Suppose that z satisfies $z\bar{z}+(1-2i)z+(1+2i)\bar{z}\le 3$.

    a. Find the equation of the locus of z and sketch this locus on an Argand diagram.

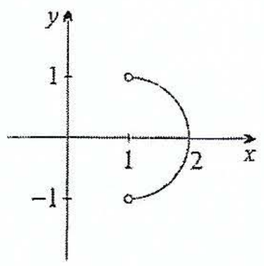
    b. Find the maximum and minimum values of $x+y$. (Hint: Consider the line $x+y=k$.)

Answers:

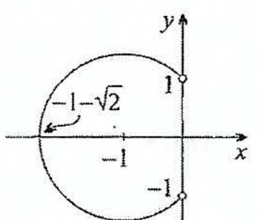
1. a.



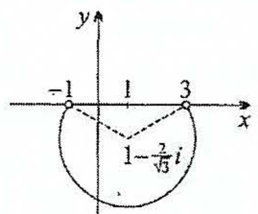
b.



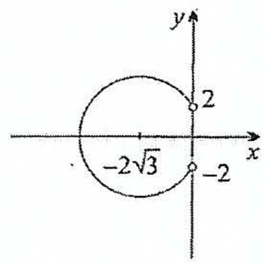
c.



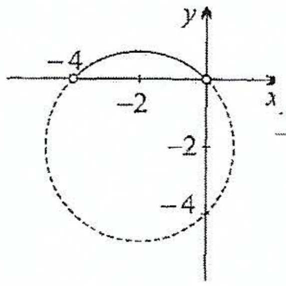
d.



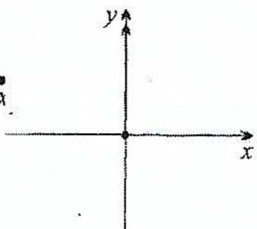
e.



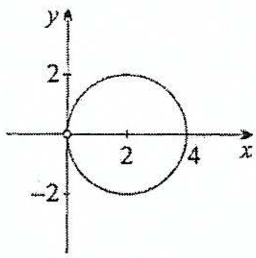
f.



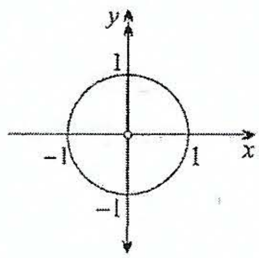
2. a.



b.



c.



3. $\sqrt{2}+2$

6. $(-1)^k\sec\left(\frac{4k+1}{4m}\pi\right)^m\cdot\frac{1+i}{\sqrt{2}}$

7. $b.\ cis\left(-\frac{3\pi}{5}\right), cis\left(-\frac{\pi}{5}\right), cis\frac{\pi}{5}, cis\frac{3\pi}{5}, -1.$

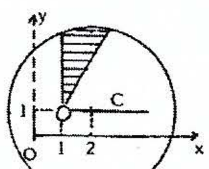
8. $a.\ \overrightarrow{AM}\ represents\ \frac{1}{2}(z_2-z_1), \overrightarrow{MP}\ represents\ \frac{i}{2}(z_2-z_1)\cot\frac{\beta}{2}$

9. $a.\ z_1=1+\sqrt{3}i, z_3=-3+\sqrt{3}i$ $\quad c.\ i.\ \theta=0°, ii.\ \theta=\frac{\pi}{2},\frac{3\pi}{2}$

11. a. i. $\frac{\sqrt{13}}{5}, \theta=-70°34'$

    ii. $\frac{\sqrt{29}}{5}, \theta=-15°4'$ c. The circle $x^2+y^2=1.\ r=1, C(0,0)$

12.



13. $\tan 4\theta=1$, $\therefore$ roots $=\tan\frac{\pi}{16}, \tan\frac{5\pi}{16}, \tan\frac{9\pi}{16}, \tan\frac{13\pi}{16}$

14. $(x-4)^2+y^2=25$, i.e. circle centre $(4,0)$, radius 5

15. a. $(x+1)^2+(y+2)^2\le 8$; b. Maximum value is 1 and Minimum value is -7